

# PROJECT BASED EXPERIMENTAL LEARNING

COURSE – NoSQL / MongoDB

## PRACTICAL FILE

PRACTICAL NO. 1

### INSTALLATION AND CONFIGURATION OF MONGODB

SUBMITTED TO – \_\_\_MR HARENDRA SINGH RAJPOOT\_\_\_\_\_

SUBMITTED BY – \_\_\_MOHD ASHAR ABBAS\_\_\_\_\_

BCA – Data Science & Artificial Intelligence

SEMESTER – \_\_\_3<sup>RD</sup> (ODD)\_\_\_ ROLL NO – 1250258270 (28)\_\_\_\_\_

UNIVERSITY / COLLEGE – \_\_\_BBD UNIVERSITY\_\_\_\_\_

## ACKNOWLEDGMENT

I would like to express my sincere gratitude to my mentor **MR HARENDRA SINGH RAJPOOT SIR** for providing me the opportunity to perform this MongoDB practical. This experiment helped me understand MongoDB installation, configuration, server operation, MongoDB Shell connection, database and collection creation, and MongoDB Compass connectivity.

The practical was performed on a Windows system using MongoDB Community Server, MongoDB Shell (mongosh), and MongoDB Compass.

# PRACTICAL NO. 1

## Installation and Configuration of MongoDB OBJECTIVE

To install and configure MongoDB Community Server, MongoDB Compass and MongoDB Shell; verify the MongoDB server; connect using mongosh; and create a database and collection.

## REQUIREMENTS

- MongoDB Community Server
- MongoDB Compass
- MongoDB Shell (mongosh)
- Windows Command Prompt / Terminal
- Windows operating system

## THEORY

MongoDB is a NoSQL, document-oriented database management system. It stores data in flexible JSON-like documents. MongoDB organizes documents into collections, and collections belong to databases. MongoDB Compass provides a graphical interface, while mongosh provides a command-line interface for interacting with the server.

## PROCEDURE

1. Install MongoDB Community Server on Windows.
2. Verify that the MongoDB Windows service is installed and running.
3. Install/open MongoDB Compass.
4. Obtain MongoDB Shell (mongosh).
5. Connect mongosh to the local server at 127.0.0.1:27017.
6. Select/create the StudentDB database.
7. Create the students collection.
8. Verify the collection using show collections.
9. Connect MongoDB Compass to localhost:27017 and verify StudentDB.

## COMMANDS USED

```
sc query MongoDB sc qc MongoDB
mongosh use StudentDB
db.createCollection("students")
show collections
```

## IMPLEMENTATION / EXECUTION

The MongoDB service was verified from an Administrator Command Prompt and returned STATE: 4 RUNNING. MongoDB Shell connected successfully to the local server using MongoDB 8.3.11 and Mongosh 2.12.0. The StudentDB database was selected and the students collection was created successfully.

## SCREENSHOTS / EXPERIMENTAL EVIDENCE

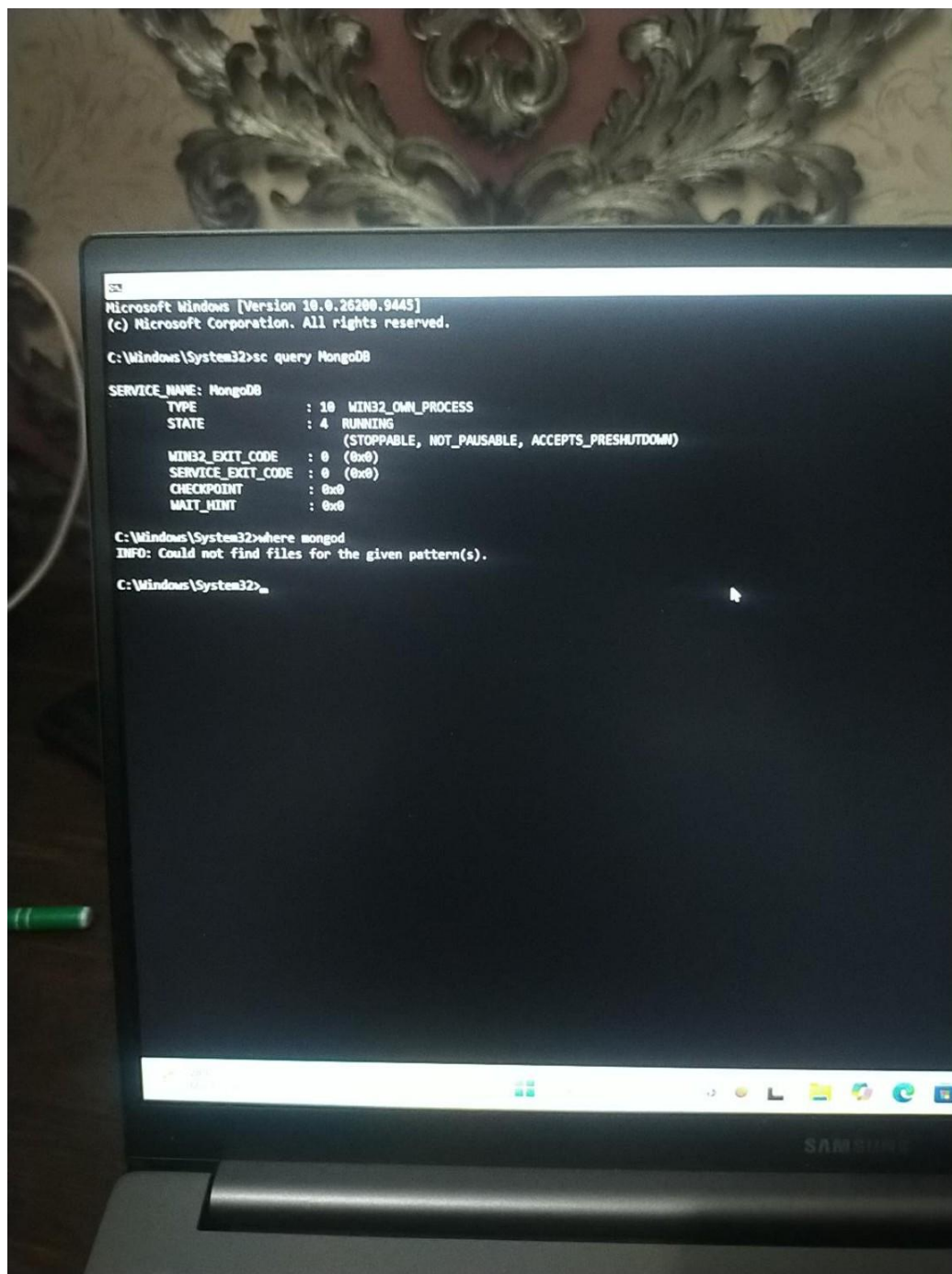


Figure 1: MongoDB Windows service confirmed as RUNNING.

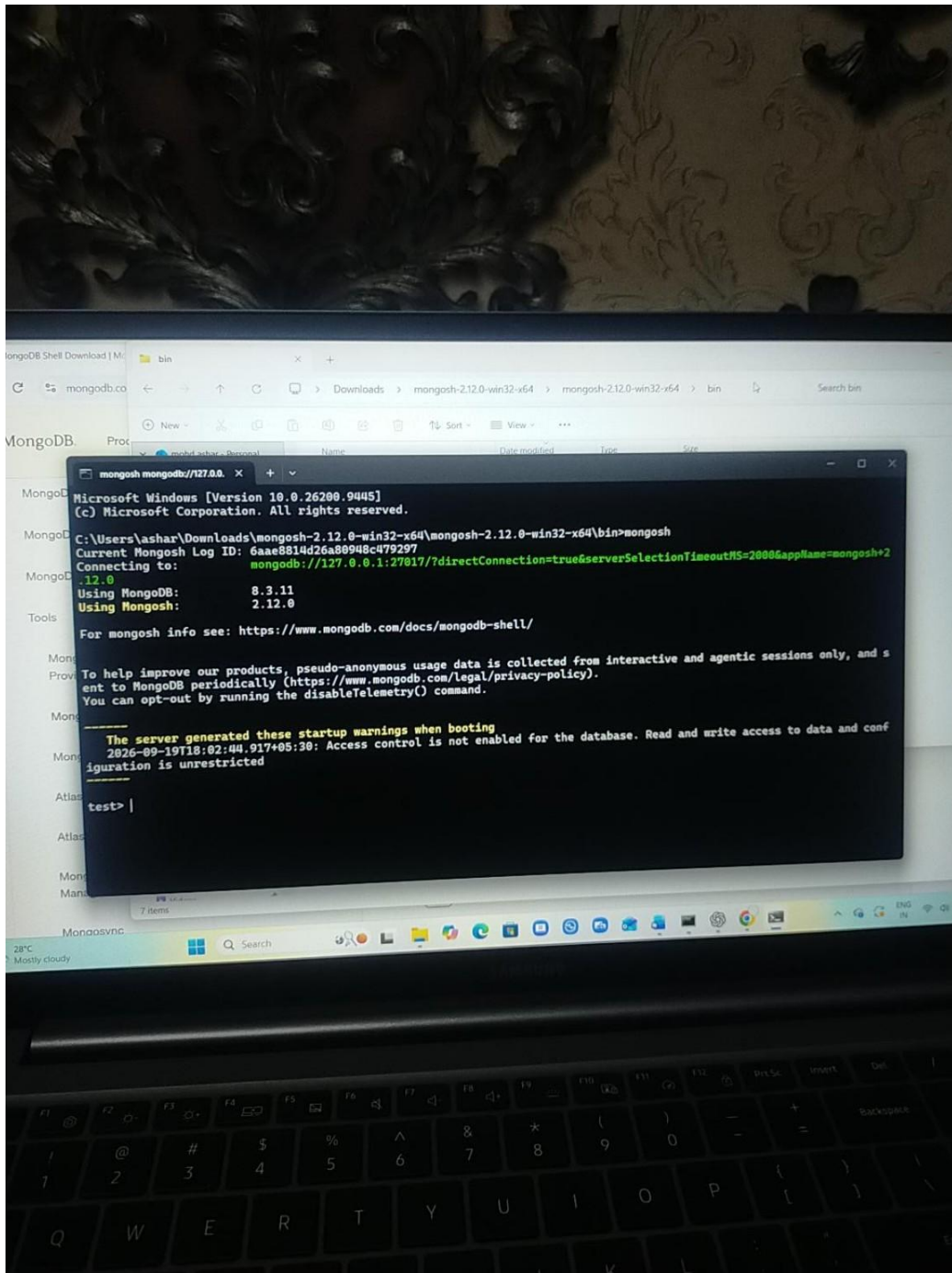


Figure 2: MongoDB Shell connected to the local MongoDB server.

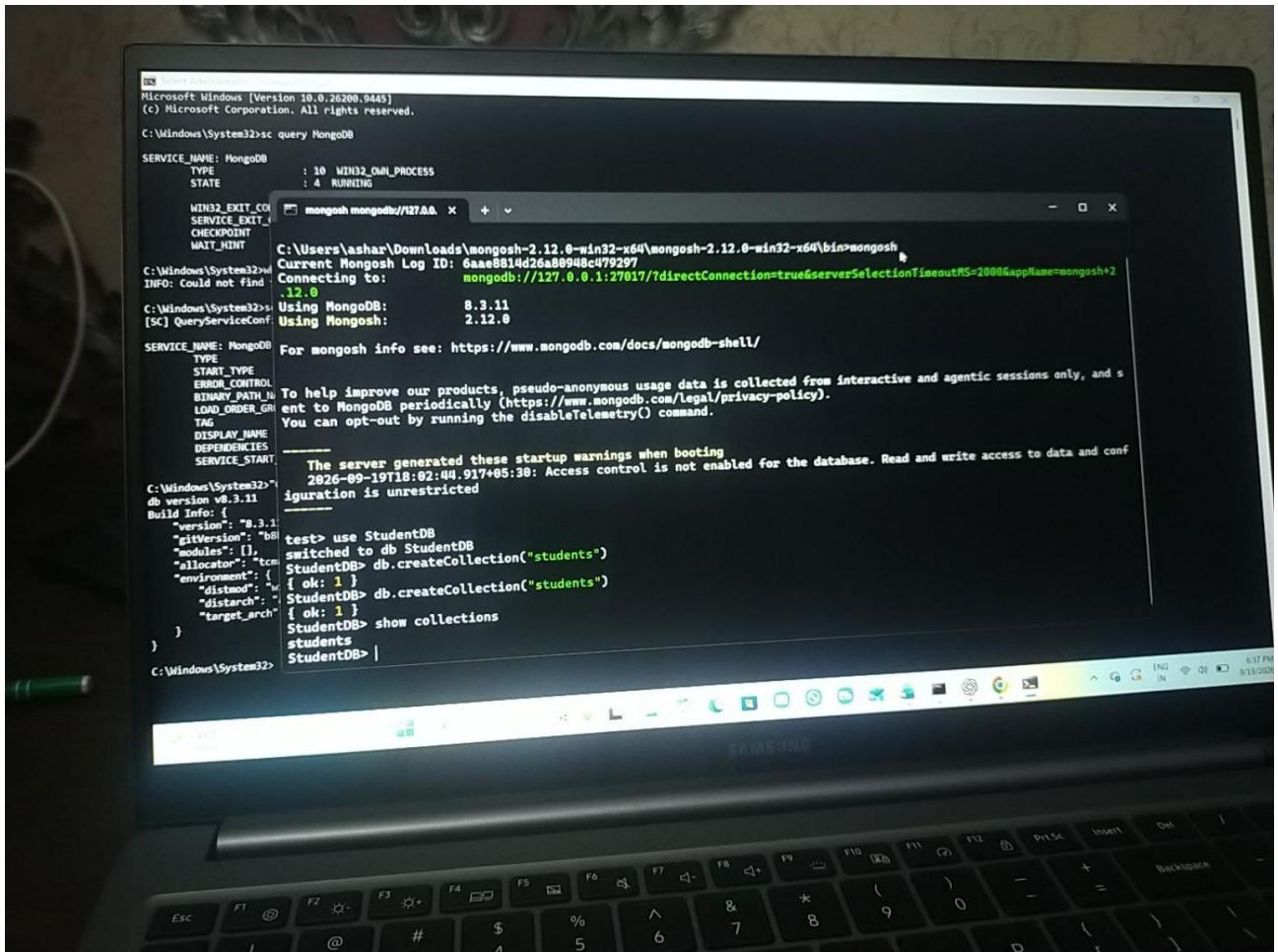


Figure 3: StudentDB and students collection created successfully using mongosh.



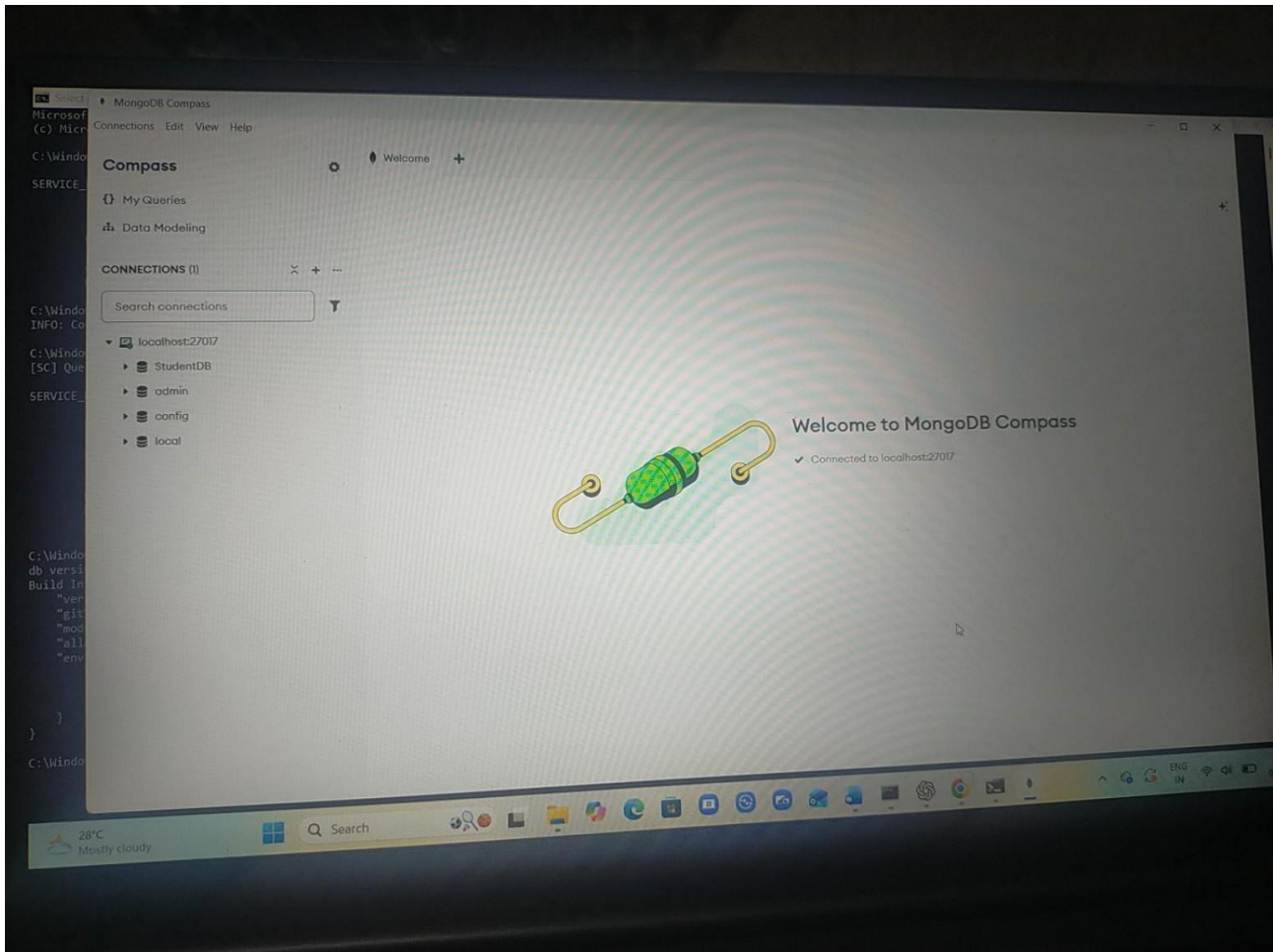


Figure 4: MongoDB Compass connected to localhost:27017 with StudentDB visible.

## RESULT

MongoDB Community Server was successfully verified as installed and running. MongoDB Shell successfully connected to the local MongoDB server. The StudentDB database and students collection were created successfully, and MongoDB Compass connected successfully to localhost:27017 and displayed StudentDB

## CONCLUSION

The practical successfully demonstrated MongoDB installation and configuration, server verification, command-line connectivity using mongosh, creation of a database and collection, and graphical connectivity through MongoDB Compass.

# THANK YOU!